

Federated Retrieval-Augmented Generation-Based LLM for Enhanced Cyber Threat Detection in the Internet-of-Energy

Tianxing Fu , Jia Hu , Geyong Min , Sunder Ali Khawaja , Keshav Singh , and Kapal Dev 

ABSTRACT

The Internet-of-Energy (IoE) represents a transformative integration of digital technologies and AI-driven analytics with energy infrastructure, creating an intelligent ecosystem that optimizes energy generation, distribution, and consumption across interconnected grids, renewable resources, and smart consumer devices. While enabling unprecedented efficiency, this interconnectivity introduces significant cybersecurity vulnerabilities, as each component presents a potential entry point for adversaries seeking to disrupt critical operations. Large language models (LLMs) have shown immense promise in addressing cybersecurity issues with their powerful natural language understanding, semantic reasoning, and robust knowledge representation capabilities. However, LLMs encounter significant limitations in processing sensitive, distributed data and executing real-time threat detection in IoE environments. In this paper, we propose FeRAG, a Federated Retrieval-Augmented Generation-based LLM system for autonomous log analysis, designed to enhance cyber threat detection performance while significantly increasing detection efficiency and mitigating privacy risks. We evaluate our method using both GPT-3.5-turbo and GPT-4o as LLM models, and our experimental results demonstrate remarkable improvements of FeRAG over other LLM-driven log analysis methods in the precision of cybersecurity threat detection. Our research also explores promising opportunities for expanding LLMs' cybersecurity capabilities through integration with diverse multi-modal data sources, enabling more comprehensive threat detection across the evolving IoE landscape.

INTRODUCTION

Recently, the Internet of Things (IoT) has experienced exponential growth over the past decade, with billions of connected devices generating vast amounts of data across diverse sectors including healthcare, manufacturing, and transportation. This proliferation has given rise to the Internet-of-Energy (IoE), a specialized IoT ecosystem that interconnects energy resources, grid

infrastructure, and consumer devices to enable intelligent energy management, optimize resource utilization, and facilitate the transition toward sustainable energy systems. As the IoE ecosystem expands, cybersecurity threats continue to escalate, given that each connected device represents a potential vulnerability that malicious attackers can exploit [1]. Large language models (LLMs) have shown immense promise in cybersecurity due to their powerful natural language understanding, semantic reasoning, and robust knowledge representation capabilities. These strengths can be leveraged in two primary ways. First, domain-specific models—such as CyBERT [2] and SecurityBERT [3]—are fine-tuned on extensive cybersecurity corpora, enabling them to capture the nuanced language of cyber threat intelligence and network traffic data, and thereby excel at tasks such as threat detection and malware classification. Second, general-purpose LLMs, exemplified by GPT-4 [4], can utilize their broad general knowledge and be guided via carefully crafted prompts or chain-of-thought reasoning to address cybersecurity challenges [5], [6].

Despite their advantages, deploying LLM-based solutions in the IoE domain poses several challenges. Firstly, general-purpose LLMs often lack the specialized knowledge required to detect emerging threats accurately, highlighting the need for fine-tuning or additional training on domain-specific data. Secondly, the intensive computational requirements of these large models, often exceeding billions of parameters, can be a major barrier in resource-limited IoE settings, where real-time responsiveness is essential. Third, even after adaptation, continuously updating these models to keep up with rapidly evolving threats imposes an additional computational burden. Lastly, handling sensitive data from the energy network raises significant privacy concerns, which requires robust security measures during deployment.

One promising approach to address these challenges is Federated Learning (FL). FL is a distributed learning paradigm that allows multiple clients (data owners) to collaboratively train a global model without sharing their raw data.

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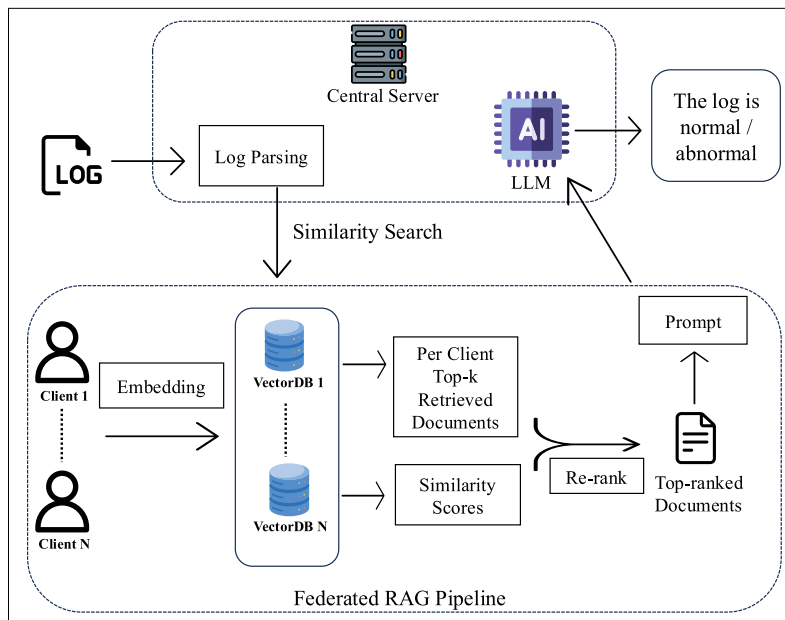


FIGURE 1. The architecture of FeRAG.

By offloading training tasks to local devices and aggregating the resulting updates on a central server, FL significantly reduces both computational burdens and privacy risks for data owners. On the other hand, Retrieval-Augmented Generation (RAG) [7] has recently emerged as an effective strategy to reduce the computational burden and mitigate the need for constant model updates in LLM-based systems. Instead of relying solely on fine-tuning to enhance the LLM with domain-specific knowledge, RAG leverages an external database to provide real-time, contextually relevant information during inference, thereby enhancing model adaptability while conserving computational resources.

Therefore, to address the aforementioned challenges, we propose FeRAG, a Federated Retrieval-Augmented Generation-based LLM system to enhance cyber threat detection in the IoE domain. FeRAG employs an advanced general-purpose LLM for automated anomaly detection in IoE device logs, identifying potential cyber threats. Our approach allows each IoE client to independently construct and maintain a local database derived from its devices' log data. Upon receiving a log entry, the LLM, located on a central server, queries these local repositories to extract the most relevant domain-specific information, thus generating the precise detection result. Leveraging distributed, independently maintained databases alongside RAG, FeRAG ensures robust data privacy while significantly reducing computational demands.

The main contributions of this work are summarized as follows:

- We propose a Federated Retrieval-Augmented Generation-based LLM system for autonomous log analysis, designed to detect anomalies in IoE devices. Our method significantly enhances the domain-specific capabilities of LLMs through selective context augmentation while preserving data privacy and increasing computational efficiency.

- We evaluated our method using both GPT-3.5-turbo and GPT-4o as general-purpose LLM components, and our experiment results revealed remarkable improvements in the precision of cyber threat detection.
- We systematically introduce the integration of Federated Learning and Retrieval-Augmented Generation techniques for cybersecurity applications, and discuss future research directions for leveraging LLMs in addressing emerging IoE security challenges, including potential extensions to real-time threat response and multi-modal attack detection.

The remainder of this paper is organized as follows. The section “[Background](#)” provides background on leveraging LLMs in cybersecurity, highlighting their current limitations, and introduces essential concepts of FL and RAG. The section “[Federated Retrieval Augmented Generation \(FeRAG\)](#)” presents the proposed FeRAG framework, detailing the integration of federated retrieval mechanisms with LLMs. The section “[Simulation Results](#)” describes our experimental setup and presents performance evaluation results. Finally, sections “[Challenges and Future Directions](#)” and “[Conclusion](#)” conclude the paper with a discussion of ongoing challenges, potential future directions, and insights drawn from our findings.

BACKGROUND

Large language models (LLMs) represent a breakthrough in natural language processing, achieved through extensive training on massive text corpora. Based on transformer architectures [8], these models implement multi-head self-attention mechanisms that enable parallel processing of sequential data and capture long-range dependencies effectively. LLMs are pre-trained using self-supervised learning techniques—primarily masked language modeling for bidirectional models and next-token prediction for autoregressive models—that enable them to develop rich contextual representations, understand intricate linguistic structures, and model complex semantic relationships without requiring labeled data. The Bidirectional Encoder Representations from Transformers (BERT) [9] marked a significant milestone in this evolution, employing bidirectional context conditioning to achieve unprecedented performance on tasks including question answering, sentiment analysis, and text classification, establishing new state-of-the-art benchmarks across the NLP landscape. Building on this foundation, models such as ChatGPT [10] have revolutionized conversational AI by generating coherent and contextually rich dialogue. The latest generation of advanced systems, such as GPT-o1, have pushed the boundaries further by excelling in complex reasoning, nuanced instruction following, and creative content generation. Collectively, these advances highlight the potential of modern LLMs to extend their capabilities well beyond conventional NLP tasks.

LLMs FOR CYBER THREAT DETECTION

LLMs can perform diverse tasks by leveraging various data types, including natural language text, structured logs, code snippets, and metadata. In

the cybersecurity domain, this versatility allows them to integrate and analyze multiple data sources, enabling applications such as automated log analysis for anomaly detection, extraction of threat intelligence from unstructured reports, and malware classification by identifying subtle behavioral patterns [1]. By processing and correlating these varied inputs, LLMs offer a comprehensive approach to detecting and mitigating cyber threats. In this section, we briefly review two principal approaches to leveraging LLMs in cybersecurity.

Domain-Specific Models: Domain-specific models are tailored adaptations of large language models fine-tuned on specialized datasets, enabling them to excel at specific cybersecurity tasks. Their performance heavily depends on the relevance and quality of the fine-tuning data used. For instance, CyBERT [2] is a cybersecurity-focused adaptation of BERT, fine-tuned on cybersecurity corpora including security news articles, CVE records, and technical APT reports. By strategically expanding its tokenizer vocabulary with 1,000 high-importance cybersecurity terms and applying masked language modeling during training, CyBERT demonstrates superior performance in cybersecurity named entity recognition (NER), multi-class threat classification, and cybersecurity knowledge graph completion. Similarly, Ferrag et al. [3] proposed Security-Bert, a lightweight, privacy-preserving BERT with only 11 million parameters designed specifically for network-based threat detection in resource-constrained IoT environments. It leverages a novel Privacy-Preserving Fixed-Length Encoding (PPFLE) and Byte-level Byte-Pair Encoder (BBPE) tokenizer to convert raw network traffic into a textual format suitable for BERT. This enables effective real-time threat detection on resource-constrained IoT devices.

General-Purpose LLMs: In contrast to domain-specific models, general-purpose LLMs—such as GPT-4—rely on their extensive pre-trained knowledge and powerful semantic reasoning abilities to address cybersecurity tasks. These models utilize advanced prompting strategies or chain-of-thoughts rather than domain-specific fine-tuning to dynamically adapt to different contexts. Recent research has demonstrated the effectiveness of prompt-based approaches for log analysis and anomaly detection. Guan et al. [6] introduced LogLLM [6], which innovatively combines BERT for semantic extraction and Llama for classification, utilizing a specialized three-stage training procedure to optimize performance. LogLLM employs BERT to extract semantic vectors from individual log messages while utilizing Llama to classify entire log sequences as normal or anomalous. Liu et al. [5] further proposed LogPrompt, which prioritizes interpretability through a specialized prompt engineering technique called log template extraction with domain-specific knowledge. This approach incorporates domain expertise into prompt construction, guiding general-purpose LLMs to extract meaningful patterns from log data while providing human-understandable explanations for detected anomalies. He et al. [11] presented LLMeLog, an innovative approach that leverages LLMs to enrich log events before embedding them for anomaly detection. LLMeLog

addresses fundamental challenges in log representation by using structured prompts to guide LLMs in resolving semantic ambiguities and identifying anomaly tendencies in log events.

LIMITATIONS

Despite their promising capabilities, both domain-specific and general-purpose LLMs encounter significant limitations in IoE cybersecurity applications. First, the computational demands—whether from fine-tuning specialized models or performing inference with massive pre-trained models—result in considerable computation and communication overhead, particularly when updates must be disseminated across heterogeneous, resource-constrained edge devices. Second, these models face significant challenges in keeping up with the rapidly evolving landscape of cyber threats. Updating LLMs to recognize and respond to these emerging threats requires either extensive fine-tuning or frequent prompt engineering adjustments, both of which impose substantial operational and computational burdens on organizations. Moreover, while prompt-based and zero-shot approaches offer flexibility without requiring additional training, they often suffer from performance instability in specialized domains like IoE. Without access to relevant domain-specific context, general-purpose LLMs can produce inconsistent or irrelevant predictions, especially when handling nuanced system logs. Third, to perform optimally in specialized domains, LLMs require a substantial amount of data, whether for fine-tuning or as an external knowledge source. This need compels the aggregation of sensitive information from numerous IoE devices, thereby raising serious privacy concerns.

FEDERATED LEARNING (FL) AND RETRIEVAL AUGMENTED GENERATION (RAG)

FL allows IoE devices to maintain data privacy by keeping raw data on-device while still contributing to model training. In FL, each device trains a local model using its own data and only sends the resulting model updates to a central server. The server then aggregates these updates—typically through a weighted average based on the number of local samples [12]—and shares the updated global model back with the devices for further training. RAG [7] has recently emerged as a powerful approach that enhances LLM inference by dynamically incorporating external, domain-specific information.

In a typical RAG workflow, as shown in Figure 2, an input query is first encoded into a vector representation, which is subsequently used to retrieve relevant contextual information from external knowledge bases using semantic similarity measures. The retrieval process takes advantage of efficient vector database technologies—such as Facebook AI Similarity Search (FAISS) [13]—which can perform sub-second lookups across millions of documents. The embedding process typically employs specialized encoding models fine-tuned specifically for semantic similarity, such as OpenAI's text-embedding models, which provide high-quality semantic representations while maintaining reasonable computational efficiency. The retrieved context is then merged with the original

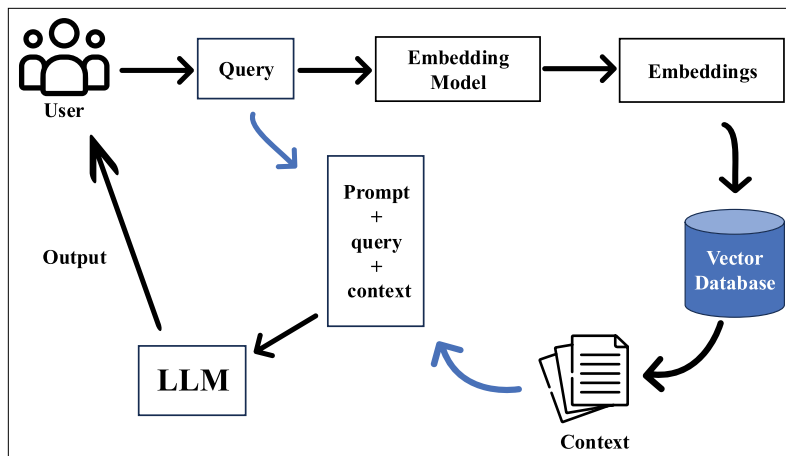


FIGURE 2. The workflow of a typical RAG system.

query to form an enriched input that the LLM processes to generate the final output. By offloading the burden of maintaining and updating domain knowledge to an external retrieval mechanism, RAG minimizes the need for expensive fine-tuning and large-scale retraining. This reduces computational overhead while ensuring that the model accesses up-to-date information on demand.

The integration of FL and RAG introduces a new paradigm for collaborative intelligence. While conventional FL relies on periodic model updates to adapt to new data, federated RAG enables the model to incorporate up-to-date, domain-specific knowledge at inference time through dynamic retrieval from distributed client knowledge bases. This approach maintains privacy by keeping raw data local while allowing the model to remain current and responsive without modifying its parameters.

FEDERATED RETRIEVAL AUGMENTED GENERATION (FeRAG)

In this section, we introduce FeRAG, a novel framework for leveraging LLMs in cyber threat detection within the IoE domain. Our approach harnesses the complementary strengths of FL and RAG to fully exploit LLMs' capabilities for cybersecurity tasks. By integrating distributed and dynamic external knowledge through vector databases, FeRAG is designed to enhance threat detection performance while significantly reducing computational burdens and mitigating privacy risks.

We consider a FL system comprising a central server (CS) that runs a powerful general-purpose LLM for anomaly detection and a network of clients. Each client is connected to multiple edge devices that continuously generate operational log data. Every client maintains its own domain-specific labeled training set that captures the distinctive characteristics of its local logs. When a new log entry arrives, the LLM leverages its reasoning abilities together with the locally learned knowledge to decide whether the entry is **normal** or **abnormal**.

Step 1 Local Knowledge Base Construction: For each client, the local training set is passed through an embedding function that converts every log entry into a fixed-length vector representation. This step captures semantic

relationships: log entries that convey similar information are mapped to vectors that lie close together when measured with cosine similarity. The resulting collection of vectors is indexed with a similarity-search algorithm so that relevant context can be retrieved quickly during inference. Each client keeps the index up to date by embedding new log data as it arrives, ensuring that anomaly detection always draws on the freshest information.

Step 2 Log Parsing and Document Retrieval:

Upon receiving a new log entry, the central LLM employs a carefully engineered prompt to perform log parsing, decomposing the input log into its static template and dynamic variable components. This parsing process generates a structured representation that emphasizes the key features and semantic content of the log entry. The structured representation is then used to query each local knowledge base, retrieving the most contextually relevant documents from every client along with their similarity scores. These results are then aggregated to form a comprehensive context set, which is subsequently passed to the CS.

Step 3 Document Re-Ranking and Anomaly Detection:

In the final step, the initial retrieval results are refined by re-ranking the aggregated documents using the reciprocal rank fusion (RRF) method. Because similarity scores from different clients may be noisy or inconsistent, this re-ranking step is essential for prioritizing the most relevant information. For each document in the aggregated context set, a relevance score is calculated by summing the reciprocal values of its adjusted ranks from all clients, where a constant is added to each rank to prevent lower-ranked items from being overly penalized. This approach ensures that documents consistently ranked highly across multiple clients receive higher overall scores. The documents are then sorted based on these scores to create a refined context set. Finally, the central LLM integrates this refined set with the original log entry into a comprehensive prompt, as shown in Figure 3, and generates the final detection result, indicating whether the log entry is classified as normal or abnormal.

SIMULATION RESULTS

In this section, we evaluate FeRAG's anomaly detection performance using a realistic dataset. We first outline our experimental methodology and implementation details.

Dataset: We use the BGL dataset [14], a large-scale log dataset collected from a Blue Gene/L supercomputer at Lawrence Livermore National Laboratory (LLNL). The dataset comprises approximately 4,747,963 log messages that have been manually labeled as either "normal" or "anomalous," with 348,460 (7.34%) of the messages identified as anomalous. From this dataset, we randomly sampled 10,000 logs and employed a stratified split, resulting in 8,000 training samples and 2,000 test samples while preserving the original anomaly distribution.

Model and Benchmark Methods: We use GPT-3.5-turbo and GPT-4o in our evaluation to

ensure a consistent comparison and to focus on the contribution of the FeRAG framework itself. In RAG systems, overall performance depends primarily on the document embedding and retrieval process, as the task mainly involves generating responses based on retrieved, relevant documents rather than requiring substantial reasoning or complex inference from the language model. As a result, models with comparable benchmark performance typically yield similar results in RAG-based applications. Therefore, We select GPT-3.5-turbo and GPT-4o because they are representative, widely used LLMs that offer stable APIs and reliable performance for our study. In addition, we include two existing log-based anomaly detection models, LogPrompt [5] and DeepLog [15], as benchmarks to provide a comprehensive assessment of our method's effectiveness. We did not include domain-specific models such as CyBERT or SecurityBERT as baselines due to their architectural designs, which are optimized for fine-tuning on labeled cybersecurity corpora. These models are not directly compatible with the RAG-based pipeline or zero-shot prompting settings employed in FeRAG.

Evaluation Metrics: We evaluate model performance using three widely-used metrics: precision, recall, and F1-score. Precision quantifies the accuracy of anomaly predictions by measuring the proportion of correctly identified anomalies among all instances flagged as anomalies. Recall assesses the model's ability to detect all actual anomalies by comparing the number of correctly identified anomalies to the total number of true anomalies. The F1-score, which is the harmonic mean of precision and recall, provides a balanced measure of the model's detection performance.

FL and RAG Setup: We construct the FL environment by simulating 30 clients, each representing a separate data silo. The training data is distributed randomly and unevenly across these clients to mimic realistic, non-identical data availability among devices. For each client, we implement the RAG pipeline using the LangChain framework, which provides modular tools for document chunking, embedding, and retrieval. Each client uses LangChain's chunking utilities to preprocess raw log files into manageable text segments and then encodes these segments into vector representations using the `text-embedding-ada-002` embedding model. The resulting embeddings are indexed locally with FAISS [13], ensuring that all log data remains private and on-device. FeRAG assumes that each client can maintain a lightweight vector database and perform real-time embedding updates, which is feasible in typical IoE deployments. For example, embedding short log entries using models like `text-embedding-ada-002` requires approximately 512–1024 bytes per vector. Maintaining a FAISS index of 10,000 such vectors consumes under 20MB of memory. For retrieval, when a new log entry is processed, the query is broadcast to all clients. Each client uses vector similarity search to retrieve the top-5 most similar document chunks from its local index and returns these results to the central server for re-ranking. We selected RRF as the reranking method due to its lightweight design and suitability for aggregating ranked results from multiple clients. Unlike more complex fusion

*RAG prompt: You are an AI log analyzer using retrieval-augmented generation (RAG) for anomaly detection.
A new log entry is provided below, along with similar logs and their classifications retrieved from clients' database, sorted by their similarities with the log entry.*

*Retrieved Similar Logs:
{retrieved contexts}*

Based on this information, classify the new log as either normal (0) or abnormal (1) using step-by-step reasoning.

*Log Entry:
{log context}*

Finally, return only the number 0 (Normal) or 1 (Abnormal).

FIGURE 3. The prompt for anomaly detection.

Model	Precision	Recall	F1 Score
LogPrompt	0.249	0.834	0.384
DeepLog	0.166	0.939	0.268
GPT-3.5 Turbo (No FeRAG)	0.158	1.000	0.272
GPT-3.5 Turbo (With FeRAG)	0.330	1.000	0.496
GPT-4o (No FeRAG)	0.120	1.000	0.214
GPT-4o (With FeRAG)	0.757	0.966	0.849

TABLE 1. Experiment results on BGL dataset.

strategies, RRF requires minimal computation and effectively emphasizes consensus among top-ranked entries, making it particularly practical for federated settings with distributed, independently ranked retrieval outputs.

Experimental Results: As shown in Table 1, the experimental results indicate that baseline models such as LogPrompt and DeepLog, along with unaugmented LLMs, tend to exhibit high recall but struggle with low precision, which points to a tendency for over-detection. For example, GPT-3.5 without FeRAG and DeepLog often flagged benign but rare patterns, such as scheduler restarts or recurring maintenance logs, as anomalies, leading to frequent false positives. In contrast, when FeRAG is integrated, both GPT-3.5 Turbo and GPT-4o demonstrate markedly improved precision, as the additional contextual information from real-time, domain-specific retrieval helps filter out these spurious alerts. The few false negatives observed under GPT-4o with FeRAG typically arose from subtle anomalies embedded in otherwise common templates, such as repeated login attempts. Notably, the FeRAG-enhanced GPT-4o shows the most robust performance overall, reflecting a better balance between detecting nearly all anomalies and minimizing incorrect classifications.

Top-*k* Selection: We conduct an ablation study to justify our choice of *k* values—the number of documents retrieved from each local database—in the RAG process. As shown in Figure 4, the F1 scores for both GPT-3.5 Turbo and GPT-4o increase notably as *k* rises from 1 to 3 and then

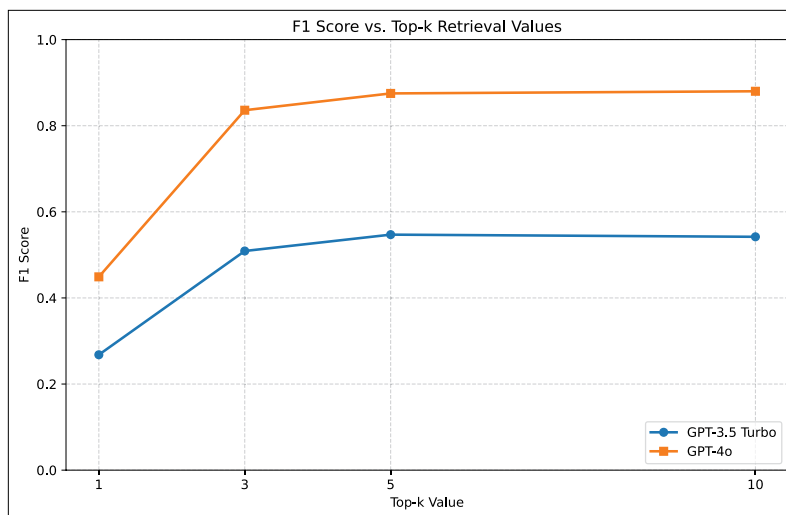


FIGURE 4. FeRAG performance with different k values.

to 5. However, further increasing k to 10 results in only marginal improvements, indicating that retrieving more than five documents offers little additional benefit and may even introduce noise. In addition, retrieving more documents results in longer input sequences for the LLM, which increases processing time and resource consumption. Therefore, we select 5 as the optimal value, as it captures sufficient context for accurate detection without introducing significant overhead. It is worth noting that the optimal value of k may vary with the size of the underlying dataset and the diversity of client logs. Larger or more heterogeneous datasets may benefit from slightly higher k values to ensure sufficient coverage of relevant variations, while smaller, more homogeneous environments might achieve optimal performance with fewer retrieved entries.

CHALLENGES AND FUTURE DIRECTIONS

Although the proposed FeRAG shows promising performance in addressing the log anomaly detection task, there are still several issues and challenges that need to be tackled in future research.

Data Heterogeneity and Embedding. Logs and telemetry from IoE devices vary significantly in format, structure, and vocabulary, complicating effective embedding and retrieval. These inconsistencies can lead to suboptimal vector representations and reduce the system's ability to surface semantically relevant context. A promising research direction is to explore adaptive embedding techniques that can dynamically adjust to inconsistencies across different IoE devices. Techniques such as domain-adaptive embedding models or context-aware chunking strategies could help create more consistent semantic representations, improving both embedding quality and retrieval accuracy. Additionally, investigating hybrid retrieval frameworks—for example, combining dense vector similarity with metadata-based or hierarchical filtering—may offer more robust handling of heterogeneity.

Multi-Modal Integration. Beyond textual logs, IoE environments generate sensor readings, categorical events, and time-series metrics, all of

which present integration challenges for LLM-based RAG systems. Directly encoding these modalities is not straightforward, as current LLMs are optimized for natural language. Research is needed on multi-modal RAG architectures capable of processing and reasoning over mixed data types—potentially by incorporating modality-specific encoders, late-fusion strategies, or multi-stream transformers—to enable more comprehensive anomaly detection.

Communication Overhead and Scalability. In federated RAG systems, transmitting large numbers of retrieved documents from distributed clients to the central server increases network traffic and computational latency, especially as context length grows. Longer input sequences for the LLM also raise processing time and resource demands. Addressing this challenge will require communication-efficient retrieval methods, such as cross-client deduplication, client-side context filtering, or embedding quantization and sparsification. Advances in LLM architectures that support efficient long-context processing, along with strategies like hierarchical summarization or sliding context windows, are essential for achieving scalable and responsive federated RAG systems.

CONCLUSION

While LLMs have shown promising capabilities in addressing cybersecurity issues, their limitations in processing sensitive, distributed data and executing real-time threat detection in IoE environments remain significant. In this paper, we introduce FeRAG, a novel framework that integrates RAG with FL to enhance log anomaly detection by leveraging distributed, domain-specific knowledge without compromising data privacy. We present a comprehensive design along with experimental results that clearly demonstrate the effectiveness of our method.

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